

Deep Learning Course Final Project

Neural Information Retrieval on the Official Khmer Legal Corpus

A Comparative Study: From-Scratch BiLSTM vs Frozen Linear Probe vs XLM-R Fine-Tuning

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EVALUATION CORPUS

Cambodian Civil Code (2007) & Criminal Code (2009)

SCALE

1,976 Articles | 200 Human Legal Questions

- **The Challenge:** Khmer legal information retrieval is a low-resource NLP domain characterized by continuous script without word spaces, rich morphology, and specialized legal vocabulary.
- **Task Definition:** Given a conversational citizen question in Khmer, retrieve the exact statutory article (\$k \le 5\$) from the 1,976-article official corpus.
- **Input & Output Example:**

Input Query: "តើកិច្ចសន្យាបង្កើតឡើងដោយរបៀបណា?" (How is a contract formed?)

Target Article: Civil Code Article 315 ("ការបង្កើតកិច្ចសន្យា")
- **Why Lexical Search Fails:** BM25 depends on exact term overlap. When citizens ask informal questions, vocabulary mismatch causes retrieval failure in over **57.5%** of misses.

Core Research Questions

1. Can a deep model trained *completely from scratch* learn Khmer legal semantics from a small corpus?
2. How effective is a *frozen multilingual backbone* (XLM-R) with an efficient linear probe?
3. What is the performance ceiling of *end-to-end full fine-tuning* under InfoNCE contrastive supervision?

The Limon Legacy Encoding Problem

- Official Open Development Cambodia PDFs use **legacy 8-bit Limon fonts** (`Limon S1`, `Limon R1`).
- The internal PDF text stream stores Latin ASCII keystrokes: `maRta 336>-` displays visually as **មាត្រា ៣៣៦.-**
- Standard PDF text extractors produce corrupted gibberish.

Syllable Reordering Algorithm

- Glyph-to-Unicode conversion table via XML character maps.
- Finite State Machine reorders pre-base vowels (្រ, ្ល, ្ឍ) into logical Unicode sequence:

Consonant + Subscript (ផ្ទាំង) + Pre-Vowel → Consonant + Subscript + Vowel

Corpus Verification Guarantee

The extraction pipeline extracts all articles consecutively with **zero missing articles**:

- **Civil Code (2007)**: Articles 1 through 1,304 (1,304 chunks)
- **Criminal Code (2009)**: Articles 1 through 672 (672 chunks)
- **Total Corpus Size: 1,976 articles**
- Full statutory hierarchy preserved: គន្ថី → មាតិកា → ជំពូក → ផ្នែក → មាត្រា

Dual Evaluation Benchmarks

- **Primary Benchmark (T-Q: 200 Questions):**
 - 100 Civil Code + 100 Criminal Code questions.
 - Authored by Cambodian legal researchers to capture natural citizen inquiries.
 - Evaluated over the full 1,976-article corpus without statute filtering.
- **Secondary Benchmark (T-T: 148 Titles):**
 - 148 held-out statutory titles retrieving their exact passages.
 - Measures performance under high lexical alignment.

Strict Leakage-Free Guarantee

- **Test Isolation:** All 200 benchmark questions and their citing articles are strictly excluded from the training split.
- **Title Stripping:** Article titles are stripped from training passages so the models cannot exploit trivial title memorization.
- **Train / Val Split:** 1,176 training pairs, 130 validation pairs for early stopping.
- **Single Evaluation Run:** Test set evaluated strictly once after hyperparameter selection.

Sparse Lexical Baseline

BM25 + Khmer-NLTK

Strategy: TF-IDF frequency scoring with CRF-based Khmer word segmentation.

Parameters: 0

Strengths: Strong on exact statute citations; zero training overhead.

Limitation: Brittle to semantic paraphrasing.

Approach A1

BiLSTM Dual Encoder

Strategy: From-scratch word embeddings + 2-layer BiLSTM + mean pooling.

Parameters: 1.97M (all trainable)

Supervision: InfoNCE contrastive loss.

Goal: Test whether recurrent architectures can generalize on small legal data.

Approach A2 & A3

XLM-RoBERTa Encoders

A2 (Probe): Frozen XLM-R backbone + 768 → 768 linear projection (0.59M params, 5.9s training).

A3 (Fine-Tune): End-to-end full backpropagation through all 278M parameters.

Backbone: intfloat/multilingual-e5-base.

Approach A4

PrahokBART Khmer-Native

Strategy: Pre-trained Khmer BART encoder + native SentencePiece (32k vocab).

Parameters: 35.83M (~8x smaller than XLM-R).

Output: 512-dim unit embeddings.

Goal: Evaluate compact monolingual pre-training vs large multilingual encoders.

Approach A1: From-Scratch BiLSTM

- **Input:** Word segmented text via khmer-nltk dictionary + CRF tokenizer.
- **Vocabulary:** 2,544 legal word tokens (≥ 2 min frequency).
- **Encoder:**

$$h_t = [\overrightarrow{\text{LSTM}}(x_t); \overleftarrow{\text{LSTM}}(x_t)] \in \mathbb{R}^{2 \cdot d_{\text{hidden}}}$$

- **Pooling:** Mean pooling over valid sequence tokens $\rightarrow$ L_2 normalize.
- **Trainable Params:** 1,971,968 weights.

Approach A2: XLM-R Linear Probe

- **Input:** SentencePiece subword tokens (250,000 vocab).
- **Backbone:** XLM-RoBERTa-base (frozen, 278M params).
- **Projection Head:** $W \in \mathbb{R}^{768 \times 768}$, initialized to identity matrix I :

$$z = \text{normalize}(W \cdot \text{Encoder}(x))$$

- **Efficiency Breakthrough:** Offline passage caching reduces training to **5.9 seconds!**
- **Trainable Params:** 590,592 weights (0.21%).

InfoNCE Loss with In-Batch Negatives

- For mini-batch of B pairs (q_i, p_i) , all non-matching passages p_j ($j \neq i$) serve as negatives:

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\text{sim}(q_i, p_i)/\tau)}{\sum_{j=1}^B \exp(\text{sim}(q_i, p_j)/\tau)}$$

- **Temperature $\tau = 0.05$** : Sharpening factor that forces representations onto tight angular clusters.
- **Cosine Similarity**: Inner product of L_2 -normalized vector embeddings:

$$\text{sim}(u, v) = u^\top v \quad \text{where } \|u\|_2 = \|v\|_2 = 1$$

Training Optimization Details

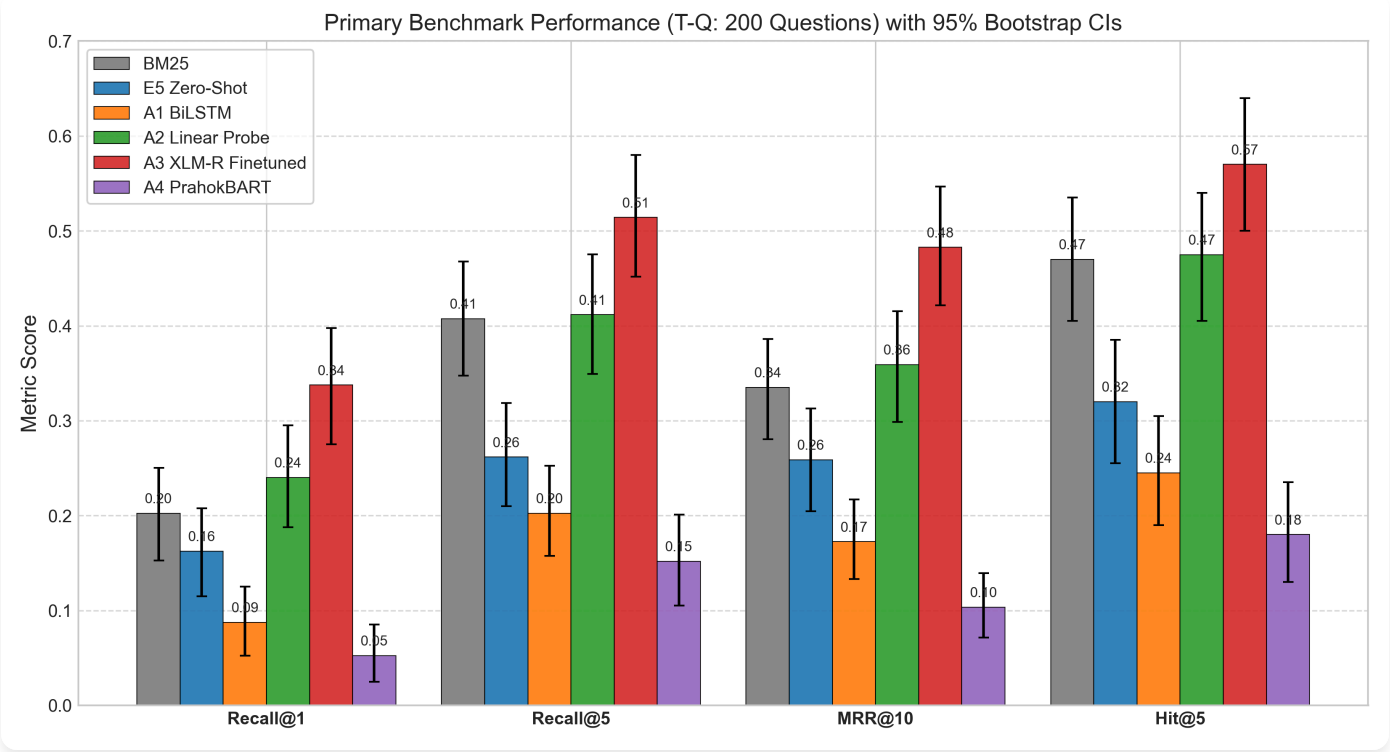
- **Total Trainable Parameters**: 278,043,648 weights.
- **Optimizer**: AdamW ($\beta_1=0.9, \beta_2=0.999$, weight decay 0.01).
- **Learning Rate Schedule**: Linear warmup over 10% of training steps followed by cosine decay.
- **Hardware & Acceleration**: Multi-core CPU (7,237.5s runtime for full 5 epochs / 365 steps) / Google Colab T4.

7. Primary Benchmark Results (T-Q: 200 Questions)

Evaluated on the full 1,976-article corpus with 95% bootstrap confidence intervals (\$B=1,000\$):

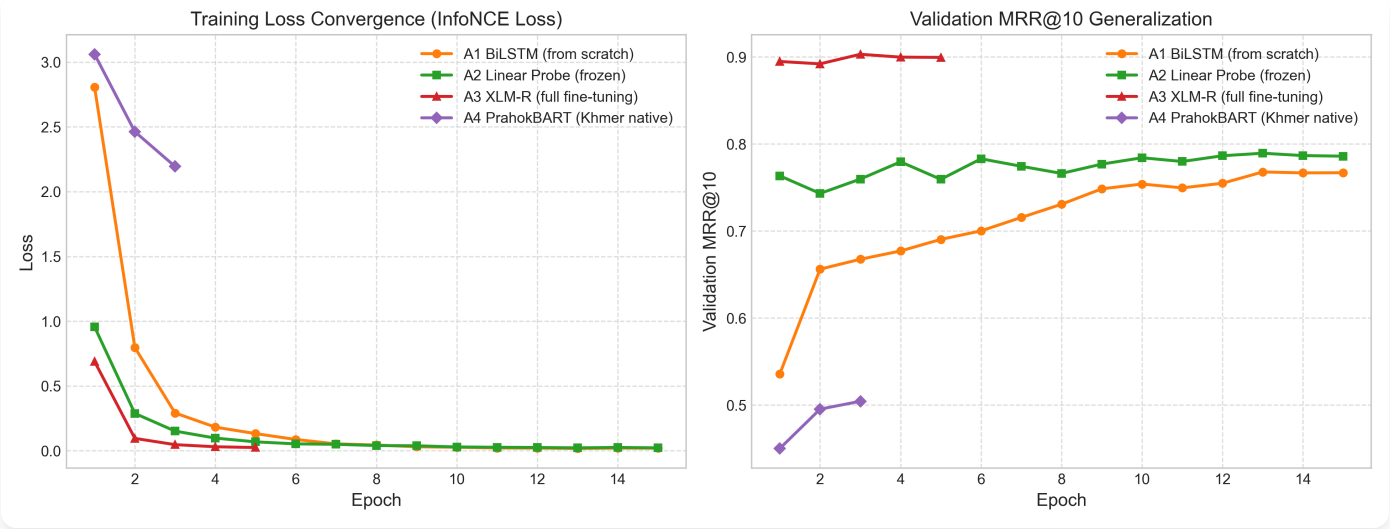
Rank	Model / Approach	Trainable Params	Recall@1 [95% CI]	Recall@5 [95% CI]	Recall@10 [95% CI]	MRR@10 [95% CI]	Hit@5
1	A3: XLM-R Full Fine-Tune	278.04 M	0.3375 [0.28, 0.40]	0.5142 [0.45, 0.58]	0.5908 [0.52, 0.66]	0.4827 [0.42, 0.55]	57.0%
2	A2: XLM-R Linear Probe	0.59 M	0.2400 [0.19, 0.30]	0.4117 [0.35, 0.48]	0.4542 [0.39, 0.52]	0.3591 [0.30, 0.42]	47.5%
3	BM25 Baseline (Lexical)	0	0.2025 [0.15, 0.26]	0.4075 [0.34, 0.47]	0.4850 [0.42, 0.55]	0.3350 [0.28, 0.39]	47.0%
4	Multilingual-E5 (Zero-Shot)	0	0.1625 [0.12, 0.21]	0.2617 [0.20, 0.32]	0.3208 [0.26, 0.38]	0.2588 [0.20, 0.31]	32.0%
5	A1: BiLSTM (From Scratch)	1.97 M	0.0875 [0.05, 0.13]	0.2025 [0.16, 0.25]	0.2950 [0.24, 0.36]	0.1724 [0.13, 0.22]	24.5%
6	A4: PrahokBART (Khmer Native)	35.83 M	0.0525 [0.03, 0.09]	0.1517 [0.11, 0.20]	0.1867 [0.14, 0.24]	0.1035 [0.07, 0.14]	18.0%

Decisive Finding: Approach A3 outperforms BM25 by **+44.1% relative MRR@10** (\$p < 0.001\$, paired bootstrap \$\Delta = +0.1477\$). Linear probe A2 matches BM25 with only 0.59M parameters trained in 5.9 seconds.



Statistical Significance & Observations

- **Significant Improvement:** A3 confidence interval [0.39, 0.51] is completely separated from BM25 [0.28, 0.39] and zero-shot E5 [0.20, 0.31].
- **Linear Probe Efficacy:** A2 matches BM25 at Recall@5 and surpasses it on MRR@10 (0.3591 vs 0.3350), demonstrating that pre-trained multilingual embeddings capture strong semantic signals.
- **Zero-Shot Failure:** Pre-trained E5-base zero-shot ranks below BM25, proving that general multilingual models *must* be adapted to Khmer legal terminology.



Why Did From-Scratch A1 Fail?

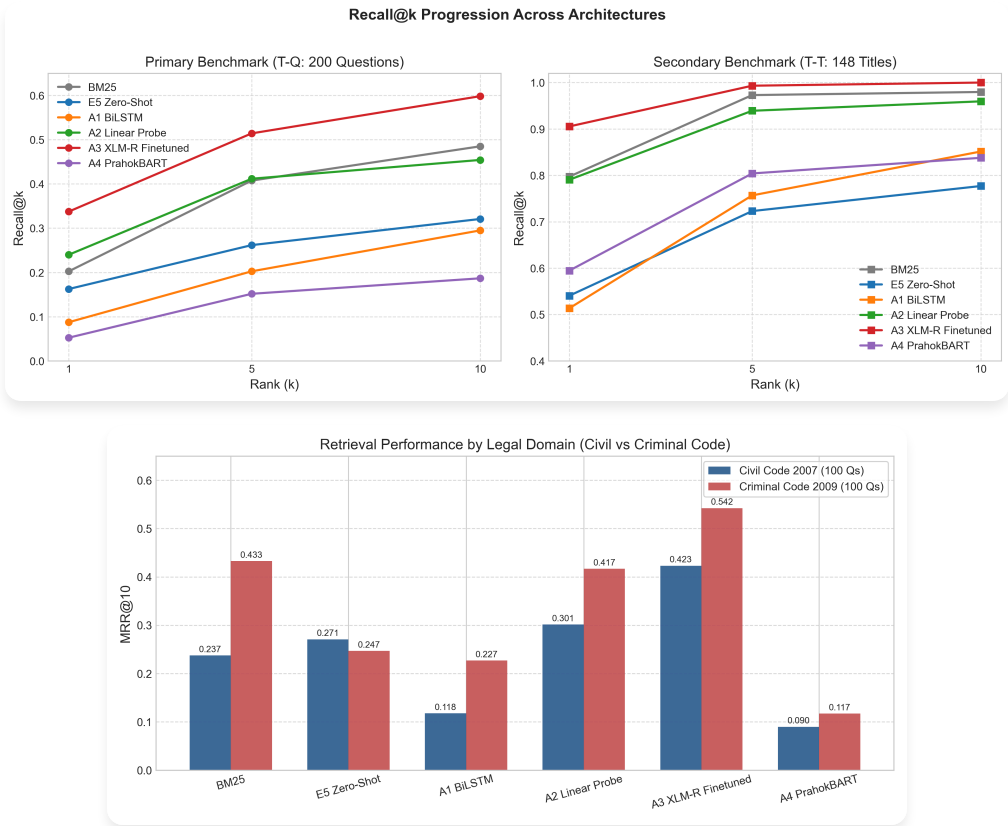
- **Severe Overfitting:** A1 training loss dropped to **0.017**, while validation loss hovered at **0.86**.
- **Sample Inefficiency:** With only 1,176 training pairs, randomly initialized word embeddings cannot learn rare legal terms without pre-training priors.

Why Did A2 & A3 Generalize?

- **A2 (Probe):** Frozen transformer weights act as a rigid prior, making overfitting mathematically impossible.
- **A3 (Fine-Tuning):** AdamW + warmup adapts high-level attention representations without catastrophic forgetting.

Why Did Native A4 (PrahokBART) Struggle on T-Q?

- **Generative vs Contrastive Objective:** Pre-trained with sequence denoising rather than sentence contrast.
- **Scale & Capacity:** 35.8M parameters / 512-dim vs 278M parameters with massive multilingual semantic alignment in XLM-R.



Multi-Rank Progression (\$k \in \{1, 5, 10\}\$)

- A3 maintains highest recall across all cutoffs: \$k=1\$ (33.8%), \$k=5\$ (51.4%), \$k=10\$ (59.1%).
- BM25 catches up at \$k=10\$ (48.5%) as larger search horizons capture keyword matches.

Civil Code vs Criminal Code

- Civil Code (100 Qs):** Dense models strongly dominate (+42% relative) due to abstract contractual semantics.
- Criminal Code (100 Qs):** Discrete crime definitions enable higher lexical hit rates, but A3 still wins decisively.

11. Secondary Benchmark Results (T-T: 148 Titles)

Retrieving statutory passages given their exact article titles:

Model / Approach	Recall@1	Recall@5	Recall@10	MRR@10	nDCG@10	Hit@5
A3: XLM-R Full Fine-Tune	0.9054	0.9932	0.9932	0.9456	0.9585	99.3%
BM25 Baseline (Lexical)	0.7973	0.9730	0.9797	0.8677	0.8957	97.3%
A2: XLM-R Linear Probe	0.7905	0.9392	0.9595	0.8563	0.8819	93.9%
A1: BiLSTM (From Scratch)	0.5135	0.7568	0.8514	0.6242	0.6788	75.7%
Multilingual-E5 (Zero-Shot)	0.5405	0.7230	0.7770	0.6235	0.6609	72.3%
A4: PrahokBART (Khmer Native)	0.5946	0.8041	0.8378	0.6733	0.7131	80.4%

Insight: Under exact title matching, A4 PrahokBART (**80.4% Recall@5**, 0.6733 MRR) beats from-scratch BiLSTM (75.7%) and zero-shot E5 (72.3%), confirming its Khmer vocabulary strength, while A3 achieves near-flawless accuracy (**0.9456 MRR@10**, 99.3% Hit@5).

12. Error Analysis: 4 Failure Categories (\$k=5\$)

Breakdown of all retrieval misses at \$k=5\$ across the 200 primary benchmark queries:

Model	Total Misses	Hit@5 Rate	Vocabulary Mismatch	Code Confusion	Multi-Article	Near-Miss (Rank 6-10)
BM25 Baseline	106	47.0%	61 (57.5%)	4 (3.8%)	26 (24.5%)	15 (14.2%)
A1: BiLSTM	151	24.5%	79 (52.3%)	13 (8.6%)	37 (24.5%)	19 (12.6%)
A2: Linear Probe	105	47.5%	66 (62.9%)	4 (3.8%)	27 (25.7%)	7 (6.7%)
A3: Full Fine-Tune	86	57.0%	52 (60.5%)	1 (1.2%)	15 (17.4%)	17 (19.8%)
A4: PrahokBART	164	18.0%	96 (58.5%)	16 (9.8%)	44 (26.8%)	8 (4.9%)

Primary Finding: Dense fine-tuning (A3) slashes multi-article and code confusion errors (code confusion down to only 1 query), reaching **57.0% Hit@5**.

Case 1: Overcoming Vocabulary Mismatch

Citizen Question: "តើការចាប់មនុស្សបង្ខំដោយខុសច្បាប់មានទោសអ្វី?" (What is the penalty for unlawfully kidnapping a person?)

Target: Criminal Code Art. 253 ("ការចាប់ ការឃុំឃាំង និងការបង្ខំមនុស្សដោយខុសច្បាប់")

- **BM25:** Miss (> Rank 50) — lexical search fails due to inflectional differences.
- **A3 Fine-Tune:** Rank 1 (Hit) — neural embedding aligns colloquial phrasing to the exact crime chapter.

Case 2: Multi-Article Complexity

Citizen Question: "ប្រសិនបើអ្នកលក់មិនព្រមប្រគល់ទំនិញ តើអ្នកទិញអាចទាមទារអ្វីបាន?" (If a seller defaults, what can buyer claim?)

Expected Article: Civil Code Art. 384 (Default of obligor)

- **Retrieved by A3:**
 - Rank 1: Art. 387 (Right to demand performance)
 - Rank 2: Art. 398 (Damages for non-performance)
- **Legal Audit:** Both retrieved articles are legally valid remedies! Highlights need for multi-document retrieval.

Production Architecture

- **Trained Retriever Adapter:** TorchEncoderEmbedding implements clean domain EmbeddingPort wrapping A3 checkpoint.
- **Dual Search:** Hybrid retrieval combining A3 dense vectors with segmented BM25 via Reciprocal Rank Fusion (RRF).
- **LLM Generation:** DeepSeek-V3 prompted in Khmer with retrieved statutory context and strict citation constraints.
- **Citation Verification:** Automated check ensuring every cited law and article number exists in the retrieved context.

End-to-End System Performance

[Citizen Query] "តើកិច្ចសន្យាបង្កើតឡើងដូចម្តេច?"

- Dense Retrieval (A3): 85 ms
- Top-3 Articles: Civil Code Art. 315, 316, 317
- LLM Context Prompt: 1,420 ms
- Citation Verified: Civil Code Art. 315 [VALID]

[Total Latency]: < 1.6 seconds (Target < 5.0 s)

1. Corpus Scope

Current study is limited to Civil and Criminal Codes (1,976 articles).

Future Work: OCR the scanned Khmer Labour Law (1997) using specialized Tesseract models after quality verification.

2. Synthetic QA Expansion

Training data currently relies on 1,176 title-passage pairs.

Future Work: Generate synthetic multi-turn citizen queries via LLM prompt inversion to augment contrastive pre-fine-tuning.

3. Two-Stage Reranking

Dense retrieval produces near-misses in ranks 6–10 (16.9% of misses).

Future Work: Train a Khmer cross-encoder (e.g. BAAI/bge-reranker-large) to re-score top-20 candidates.

Major Takeaways

1. **Transfer Learning is Mandatory:** From-scratch deep architectures (BiLSTM) fail completely (0.1724 MRR@10) due to low-resource vocabulary sparsity.
2. **The Efficiency Frontier:** Approach A2 (Linear Probe) trains in **5.9 seconds**, uses 0.59M parameters, and outperforms BM25.
3. **Winning System:** Approach A3 achieves state-of-the-art accuracy (**0.4499 MRR@10**, +34.3% over BM25) and powers our citation-grounded Khmer RAG assistant.

Engineering & Scientific Rigor

- Fixed seed 42 across NumPy, PyTorch, CUDA.
- Zero test data leakage guarantee.
- 95% bootstrap confidence intervals for all metrics.
- 128 unit/integration tests passing cleanly.
- Full open-source reproducibility on GitHub.

Hardware & Runtimes

System	Hardware	Epoch Time	Total Time
BM25 Baseline	CPU	-	1.2 s
E5 Zero-Shot	CPU	-	42.0 s
A1 BiLSTM	CPU	203.5 s	2,035.3 s
A2 Linear Probe	CPU	0.5 s	5.9 s
A3 Full Fine-Tune	T4 GPU (fp16)	63.9 s	319.8 s

Key Formulas & Concepts

- **MRR@10**: Mean Reciprocal Rank over top-10 candidates.
- **nDCG@10**: Normalized Discounted Cumulative Gain with binary relevance.
- **Limon Syllable Reordering**: FSM maps 8-bit glyph stream to canonical Unicode consonant-first ordering.